Massachusetts Institute of Technology

# High specific impulse ion electrospray propulsion for small nanosatellites

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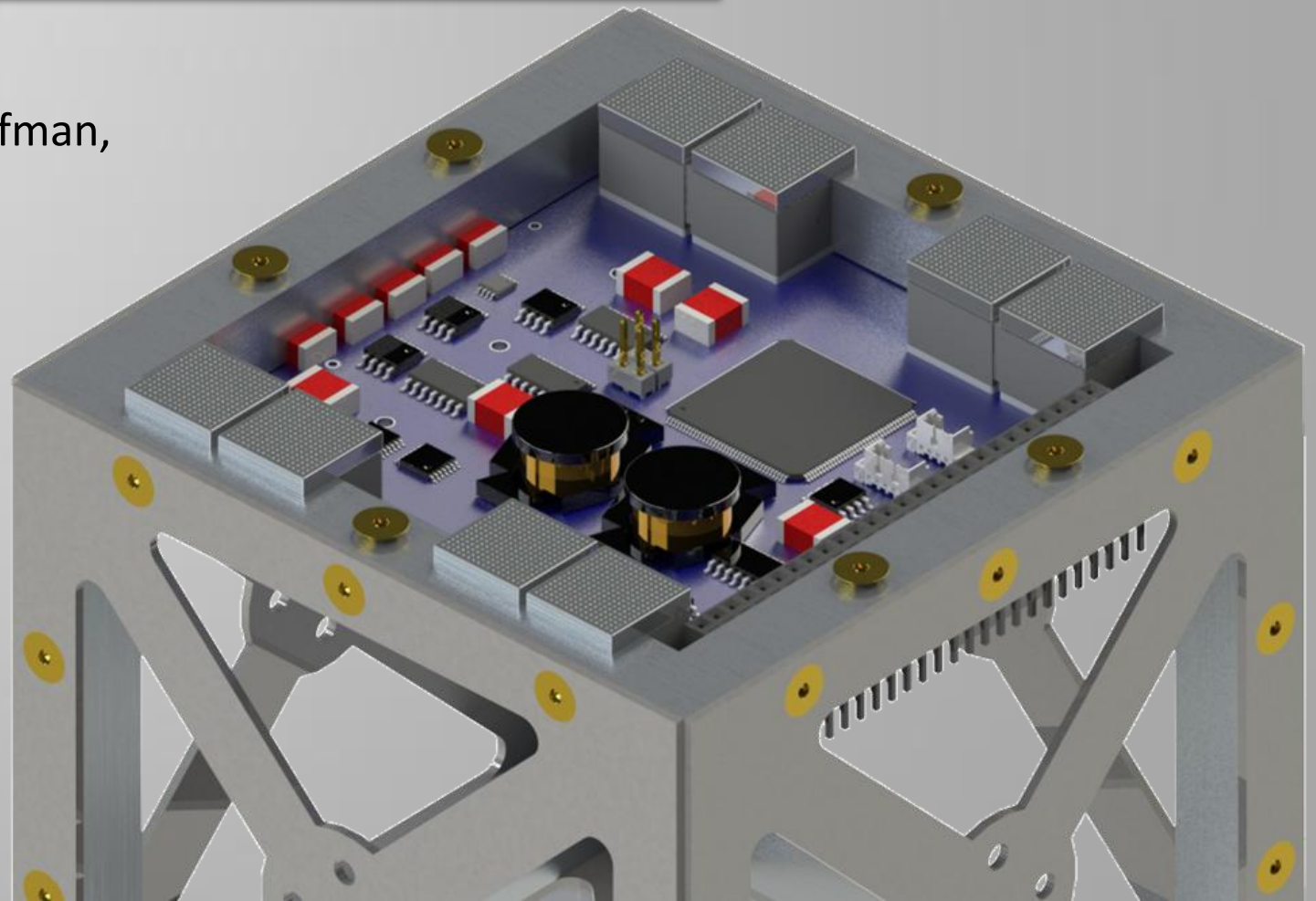
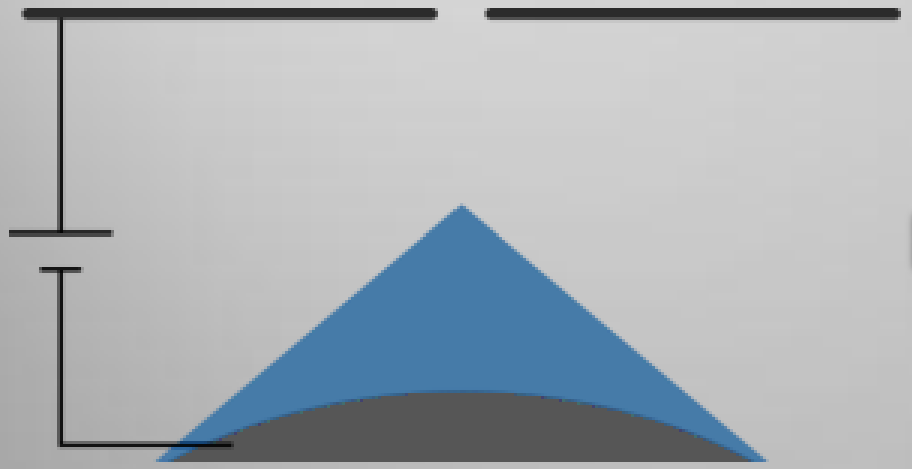


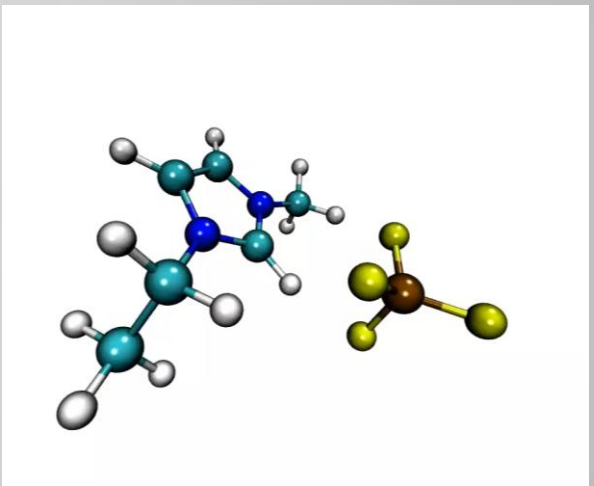
# Electrospray mechanism

1. A layer of ionic liquid rests on a porous surface

2. An electrical potential difference is created between the surface and a grid above

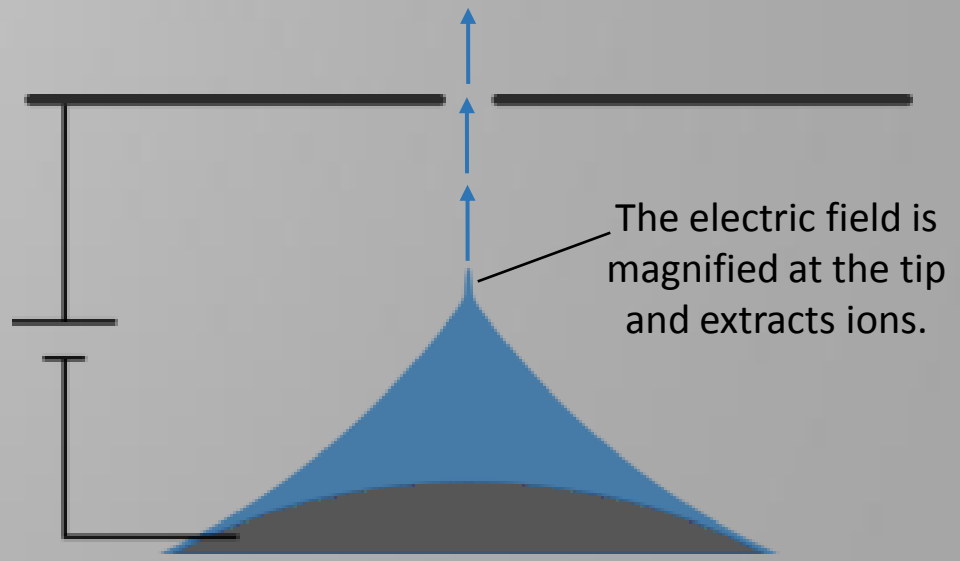


3. The liquid rises up in the electric field to form a structure called a Taylor cone



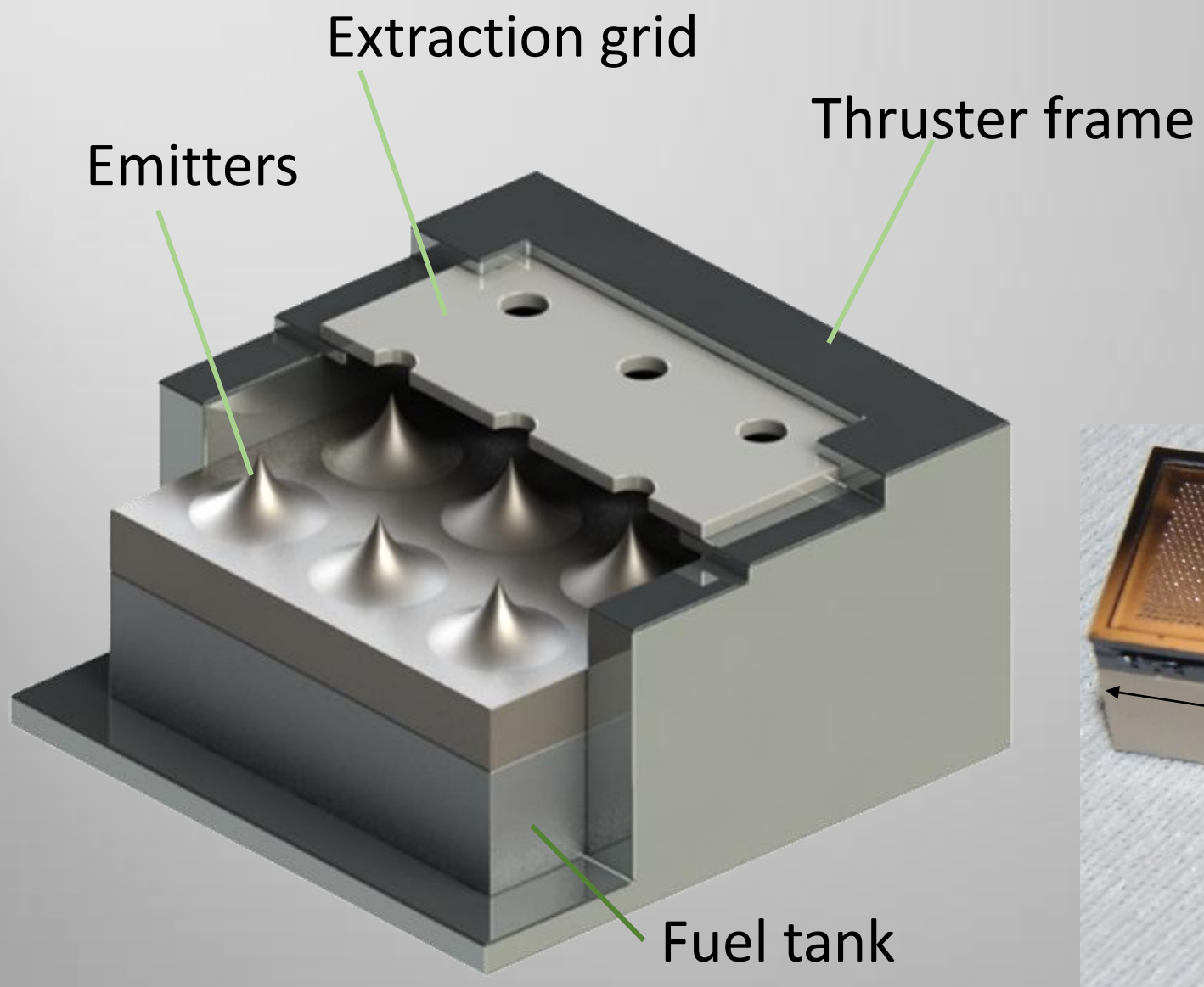
- Ionic liquid properties:
  - Plasma in a bottle, molten salt at room temperature
  - Zero vapor pressure
  - Non-Toxic

4. The electric field strength is increased

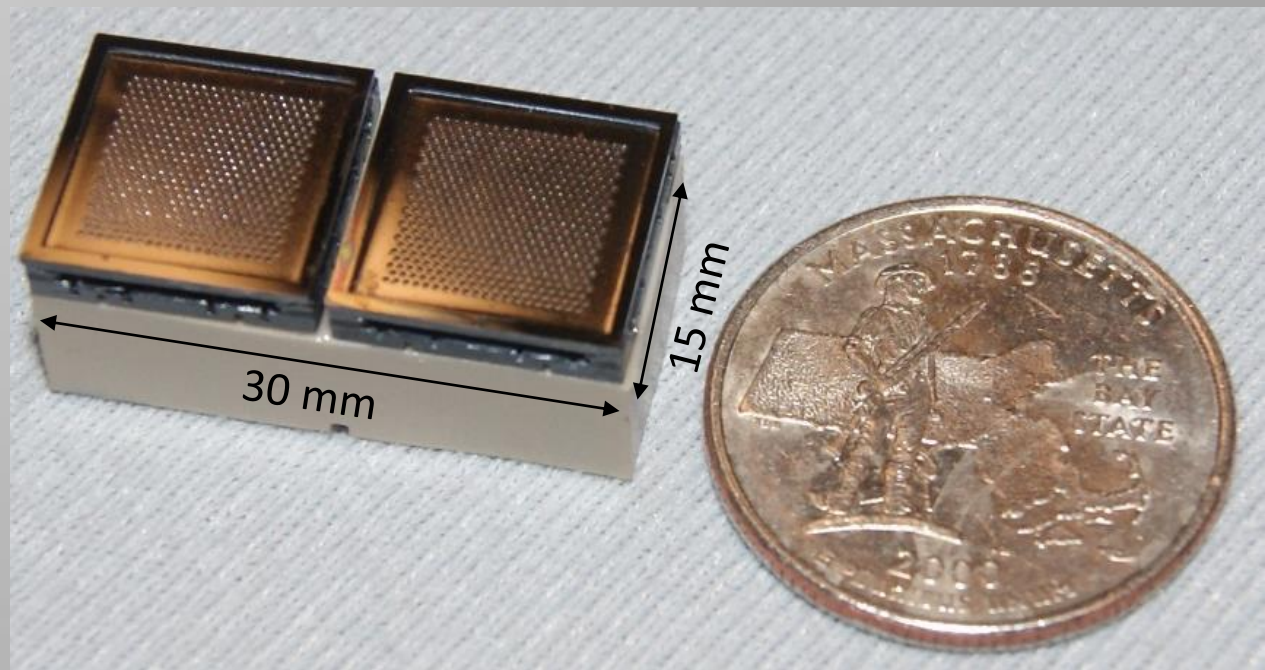



**Emitted ions escape through the gap in the grid, thereby providing thrust (40nN)**

# Electrospray array concept

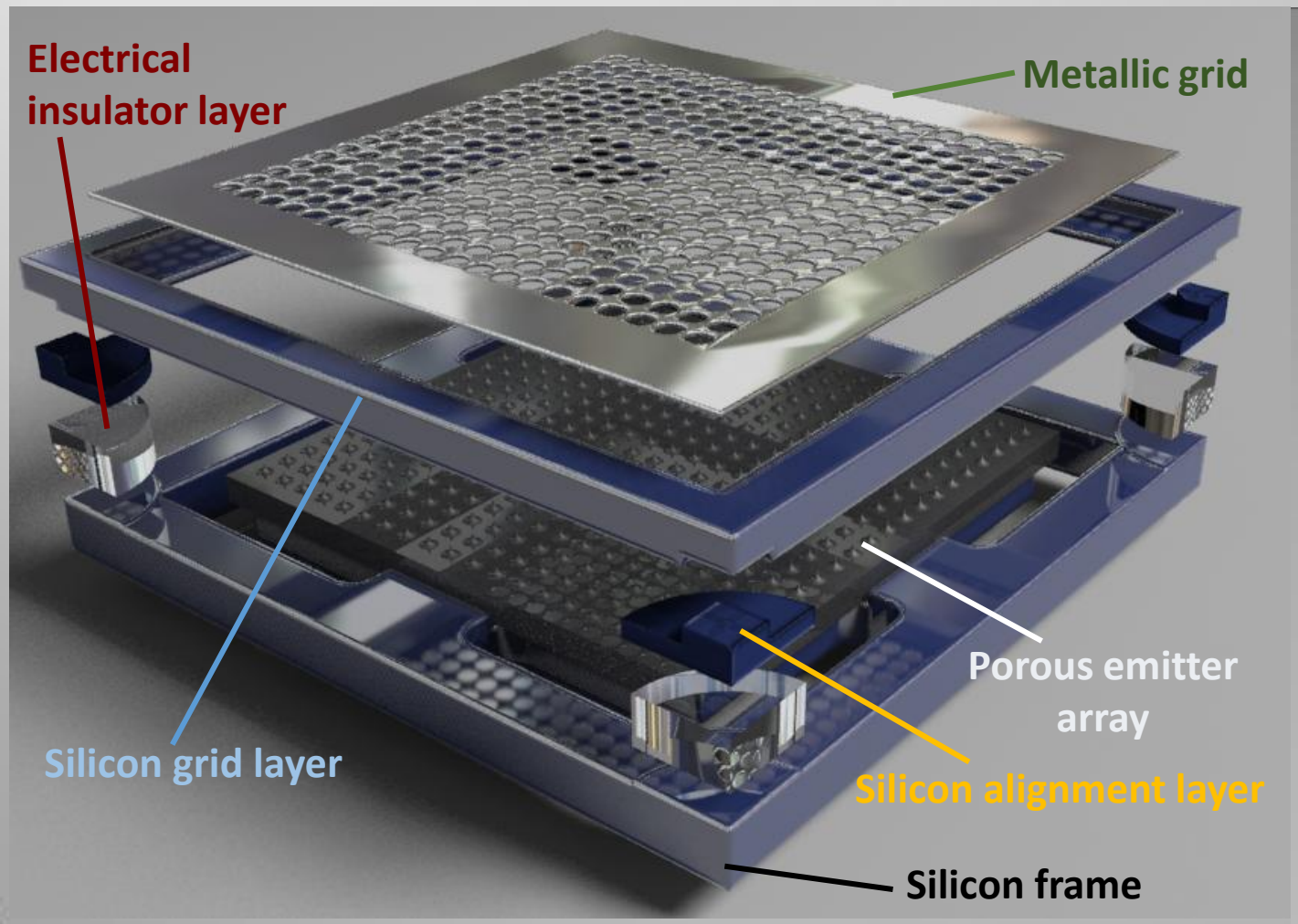


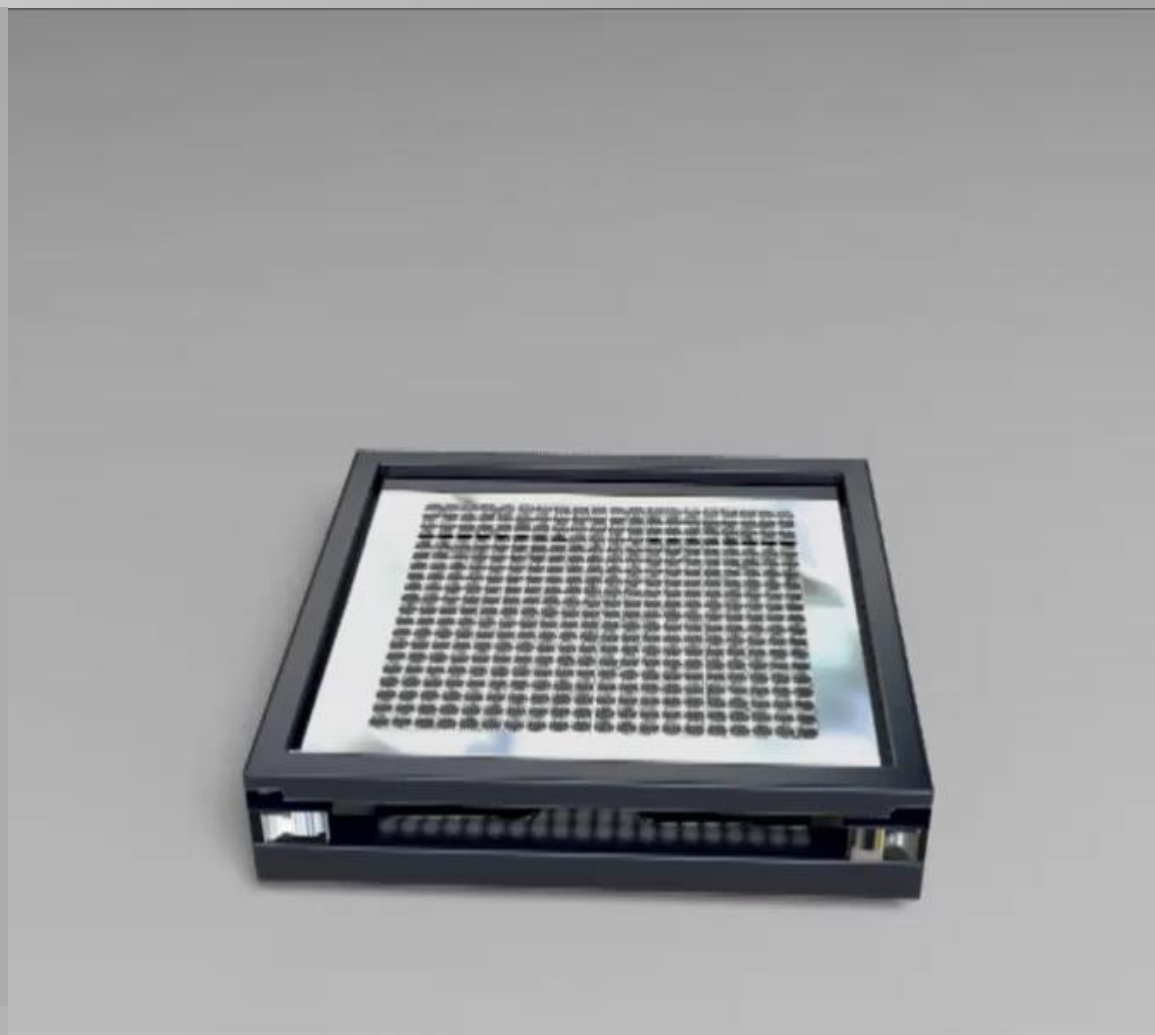

- 20 µN with 500 emitter tips
- ≈ 1 watt thruster power consumption
- System volume 100 cm^3. (thruster and electronics)
- High ISP, >2500 seconds
- Simplicity and reliability, no moving parts.

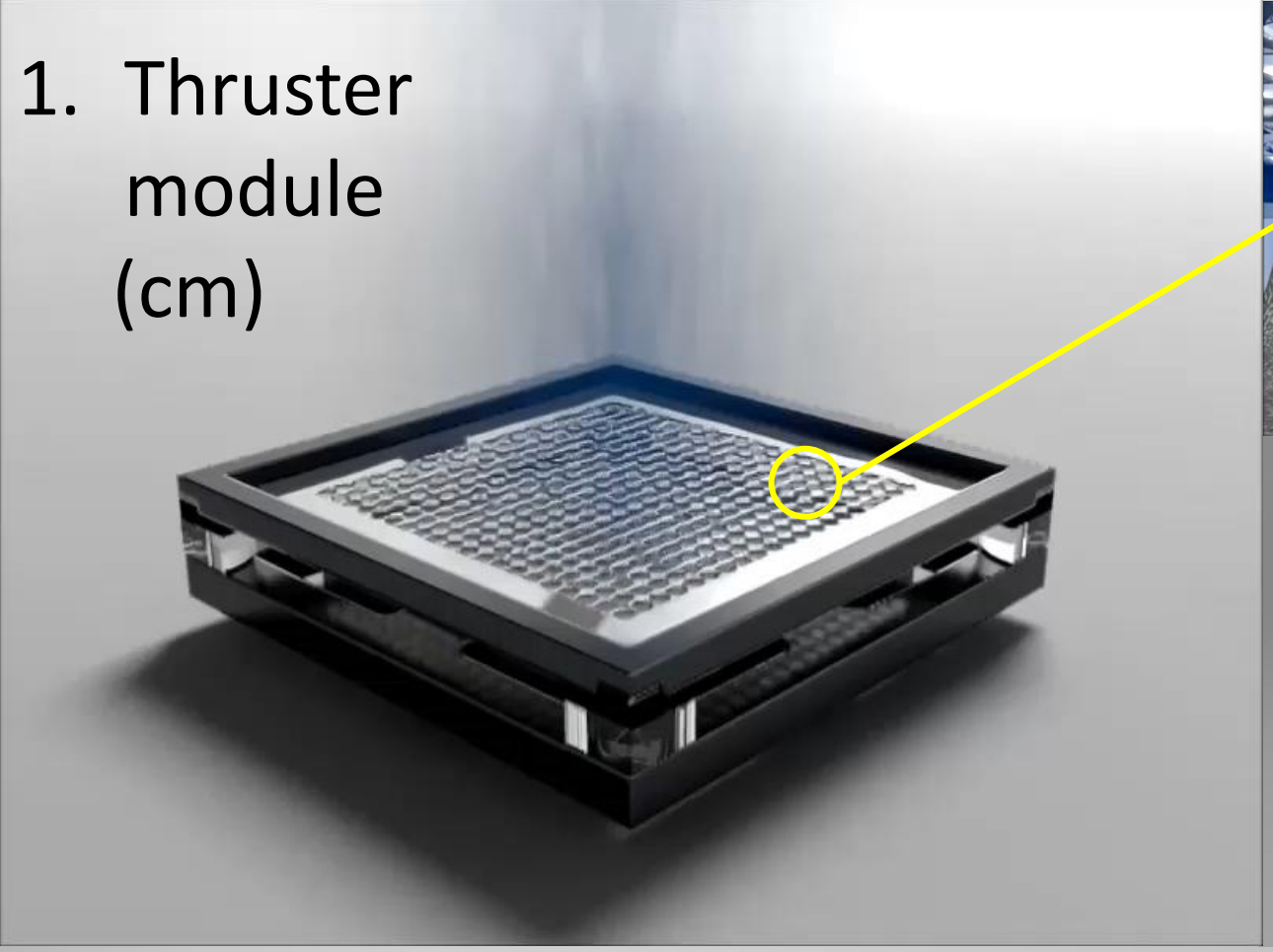


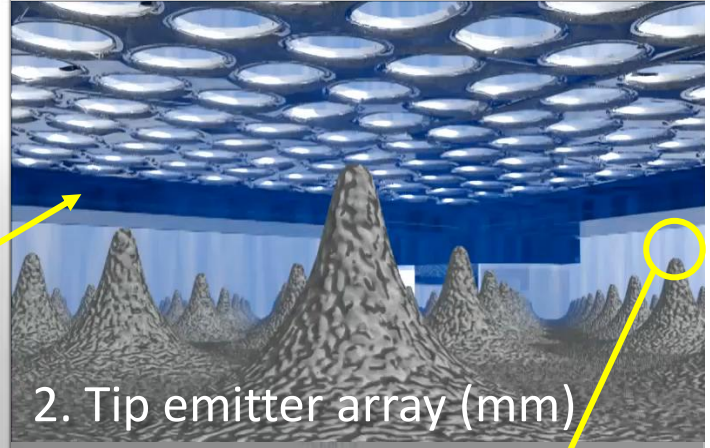
# Thruster module layers

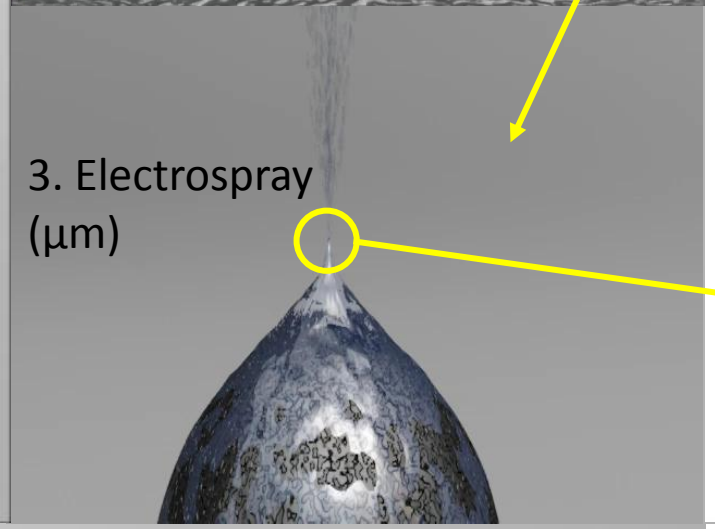
- All layers are manufactured using state of the art MEMS processes.



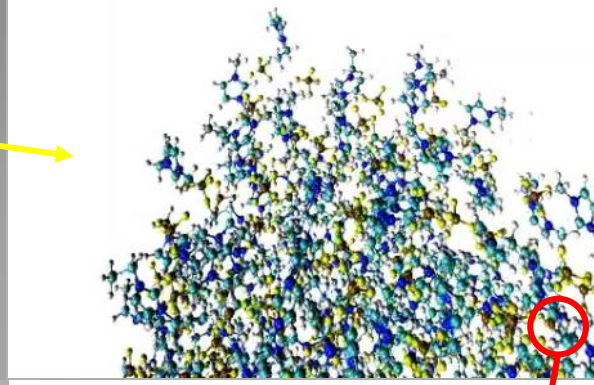





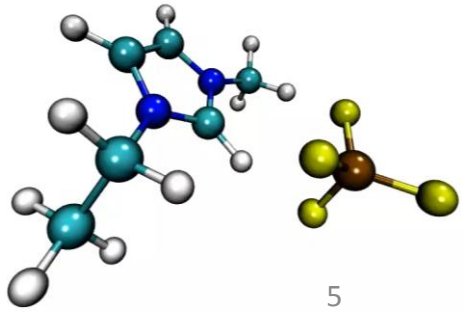




4. Ion evaporation (nm)



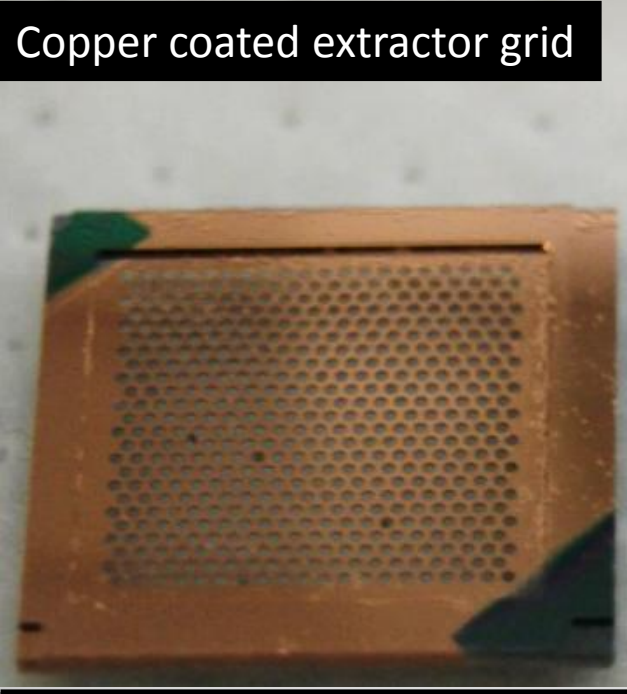
4. Ionic liquid molecule (**Å**)

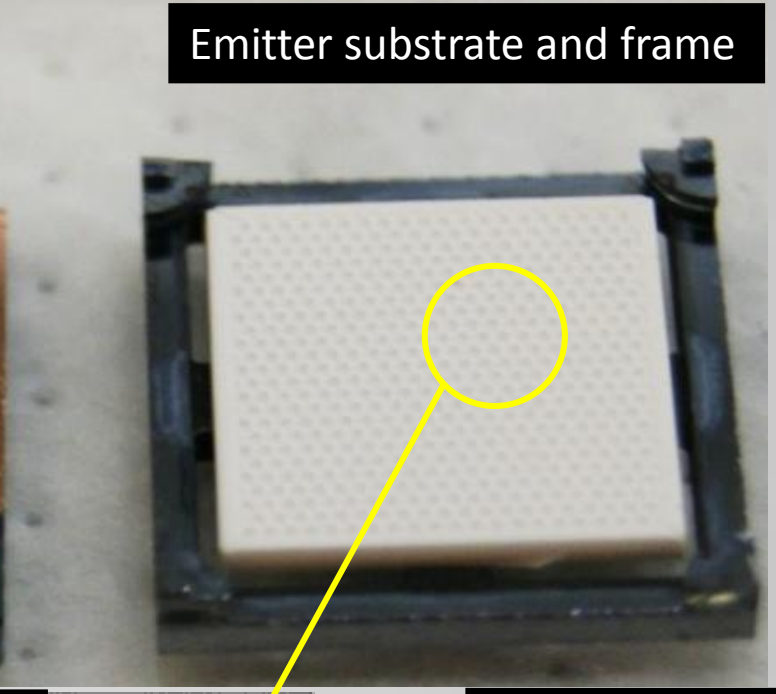


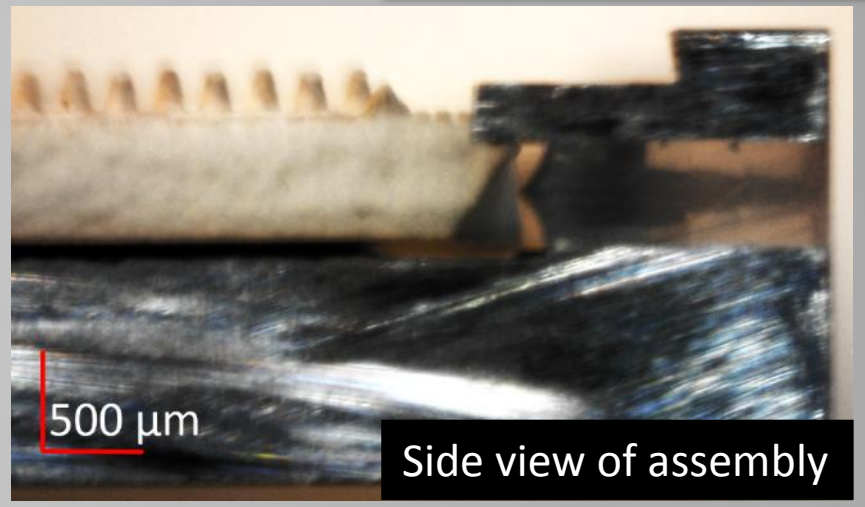
Artist conception of different levels of the electrospray module in operation. Ion plume is not visible in regular operation.

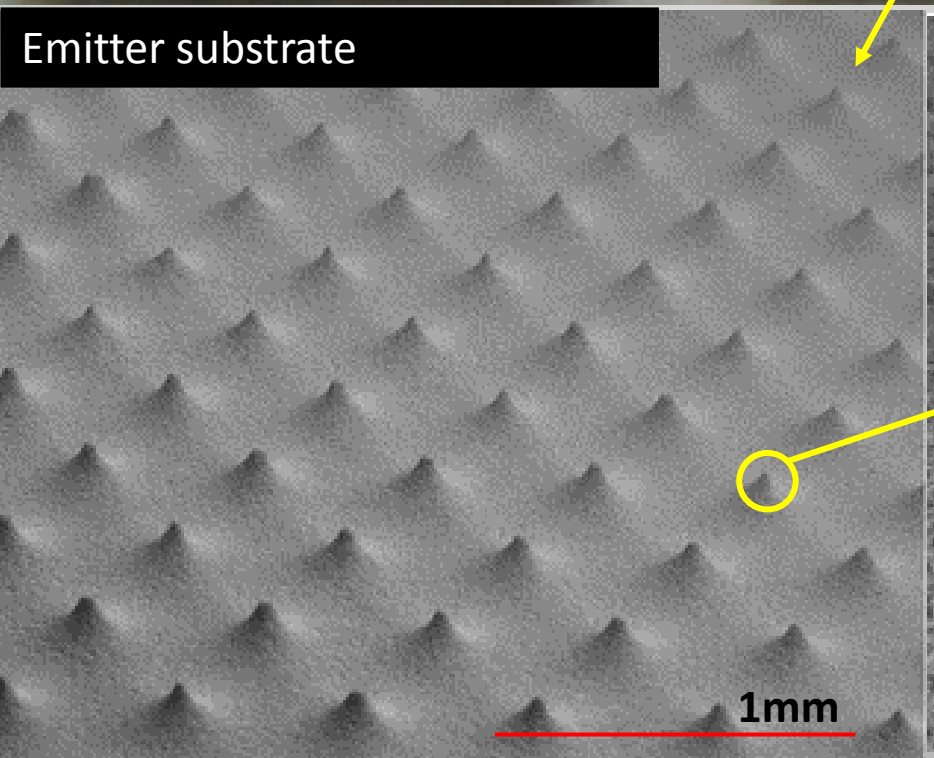
# Macroscopic to microscopic scale

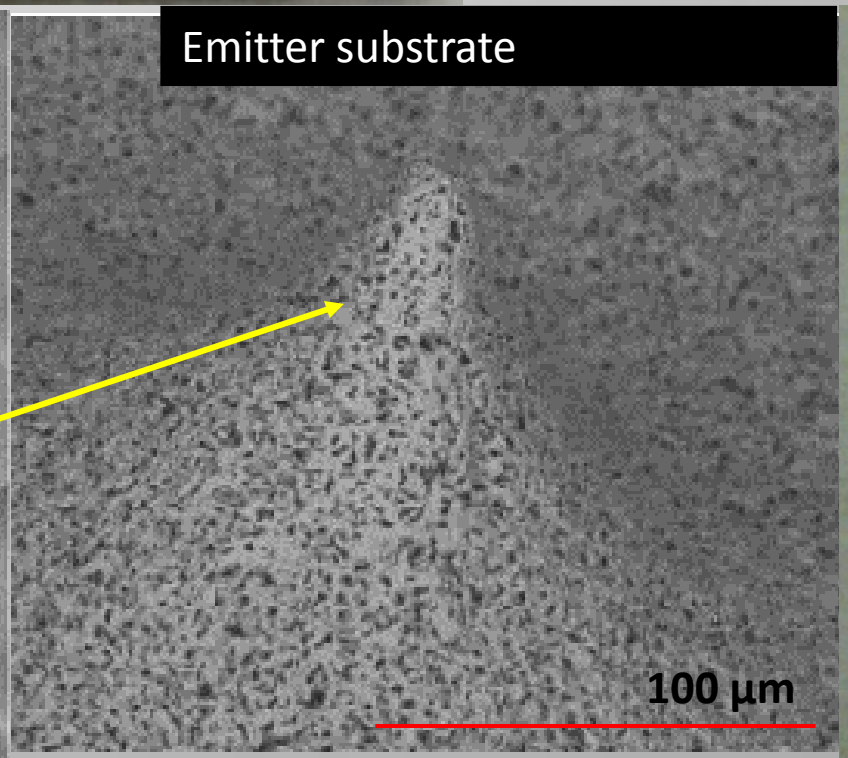
# Device details

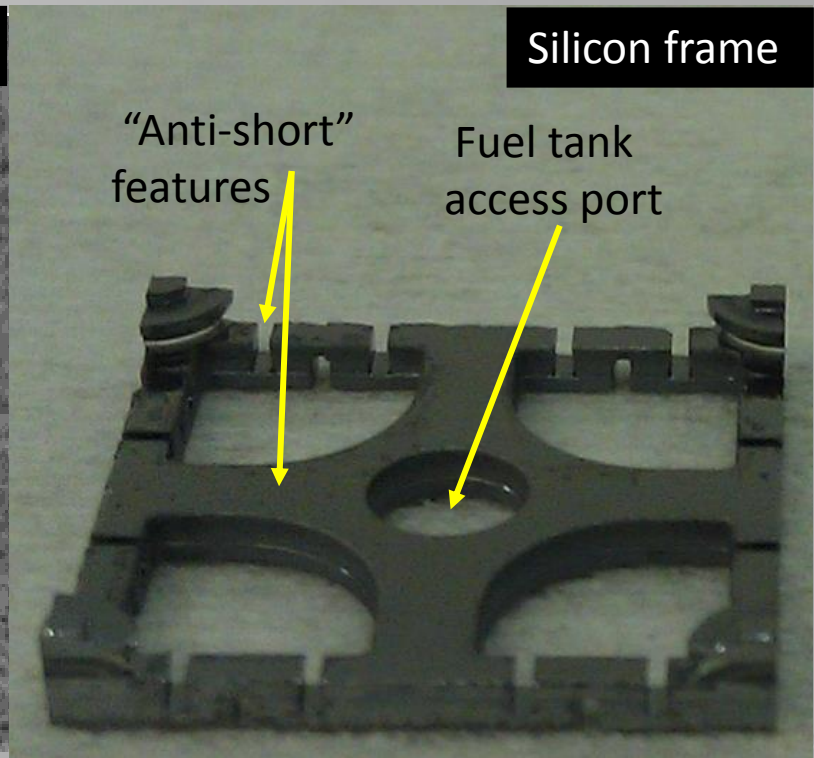


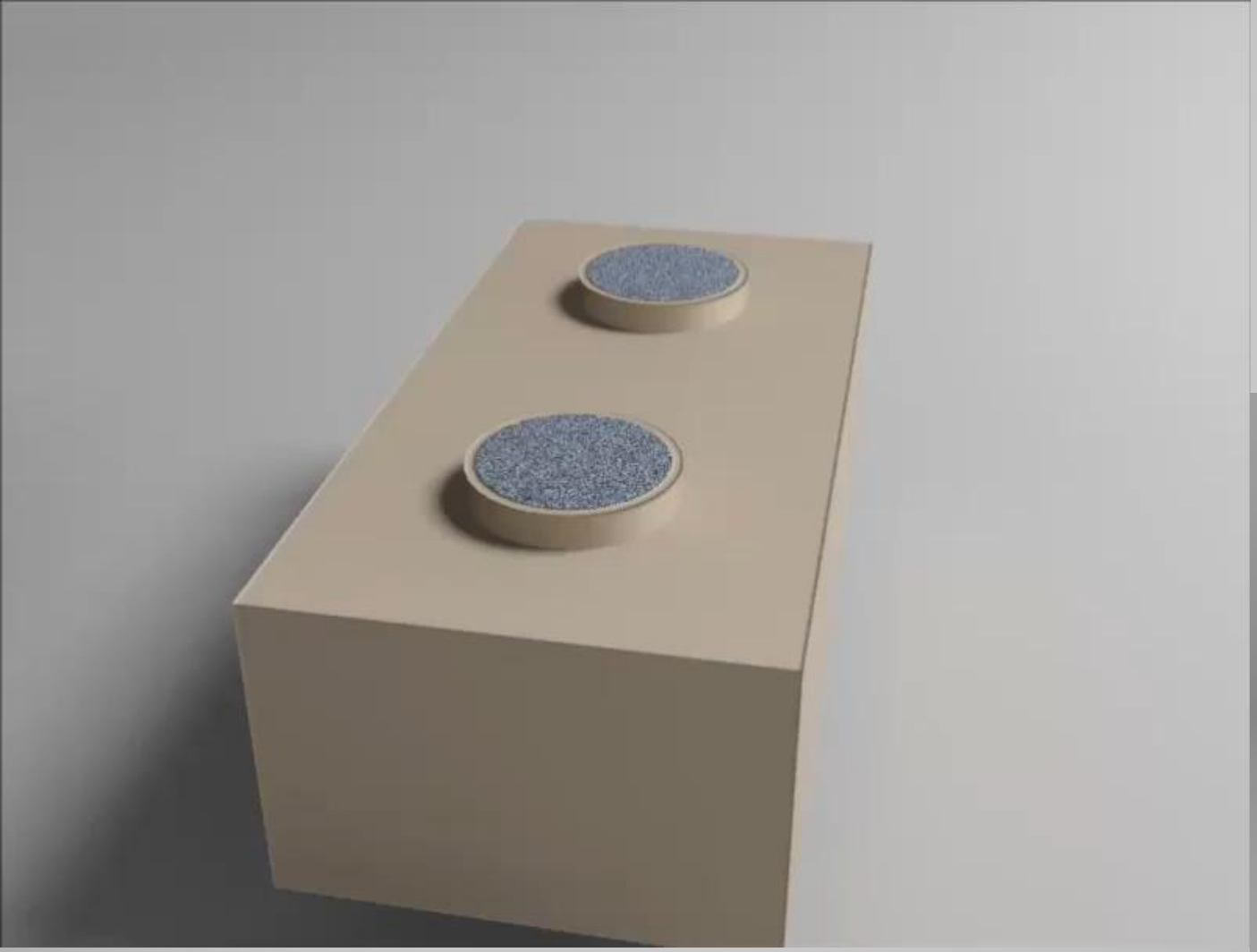



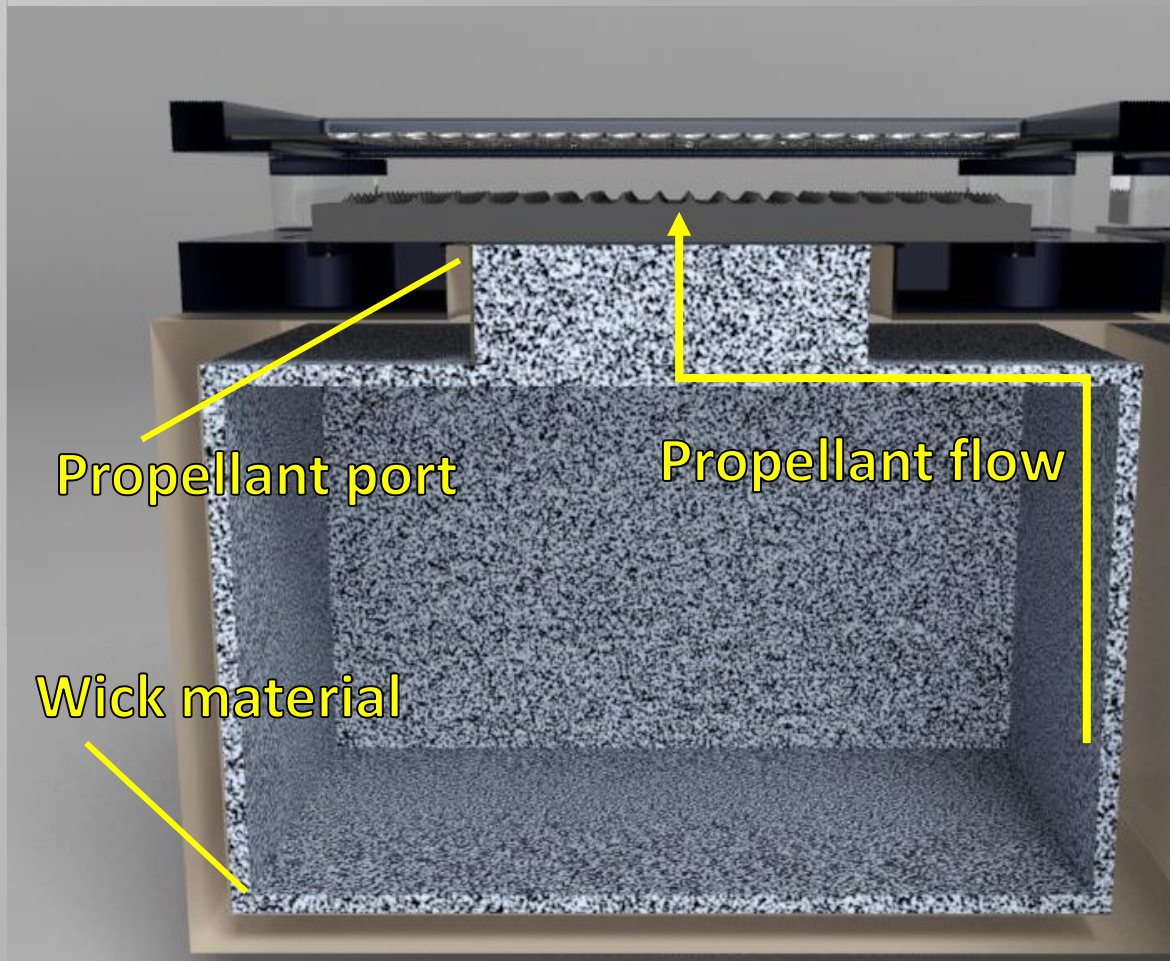



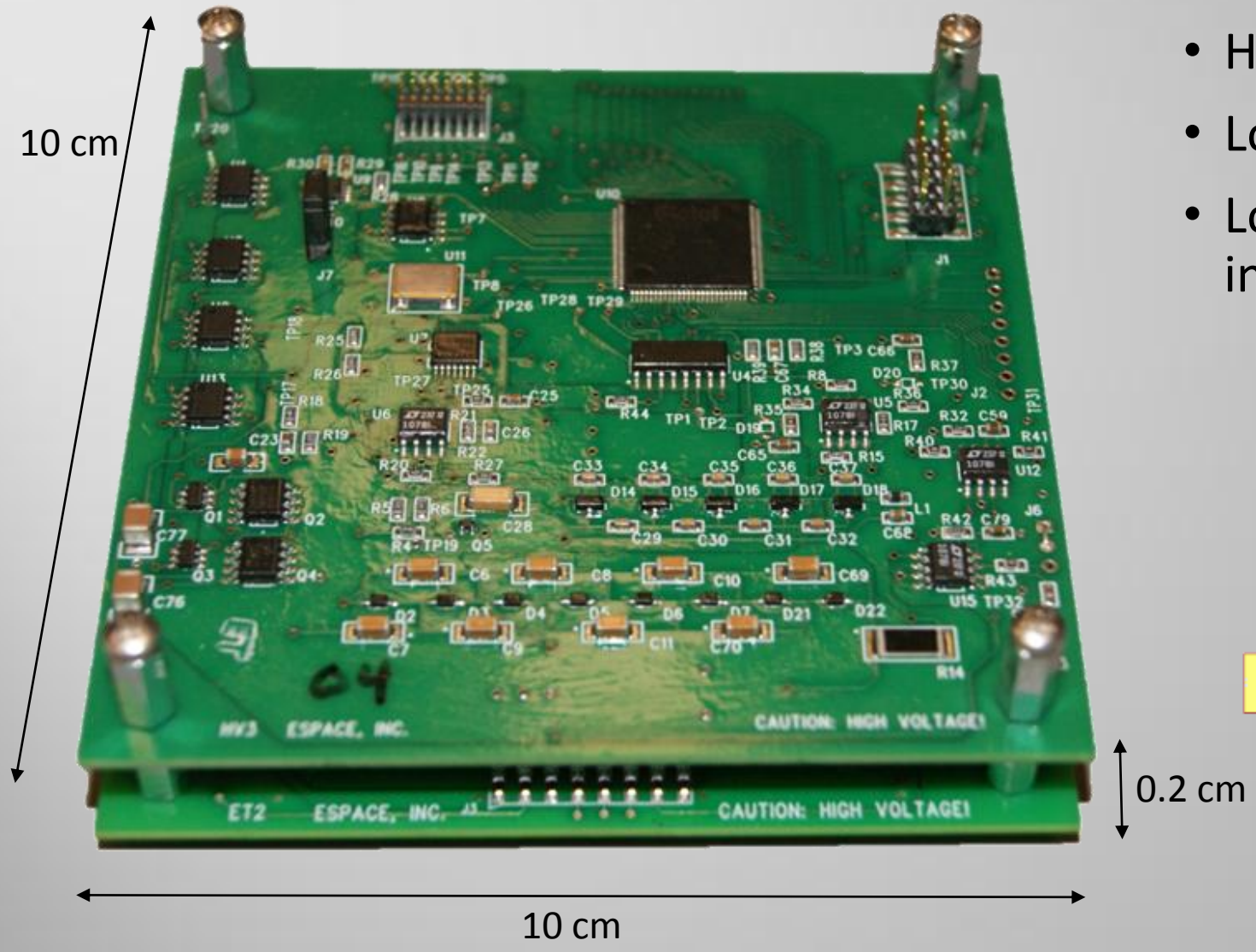
# Tank design

- Zero vapor pressure propellant.
  No valves or pressurization.

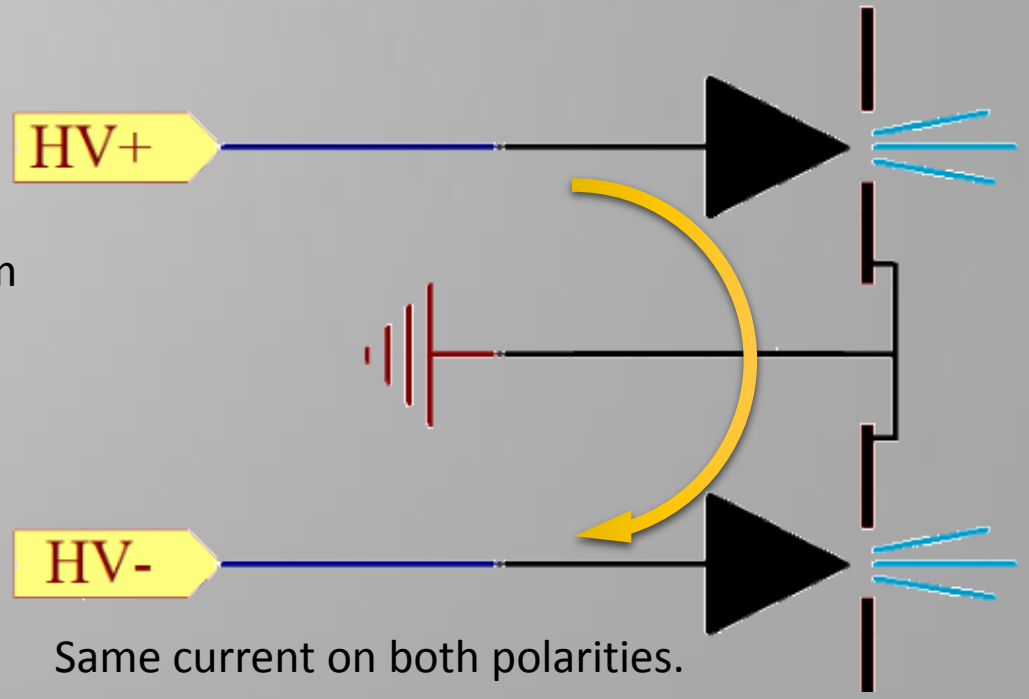





- Pore size has a gradient towards the emitters.
- Wick material has a bigger pore size than the emitter substrate.




- High voltage polarity switching
- Low power consumption
- Low System volume. Thruster integration on electronics board



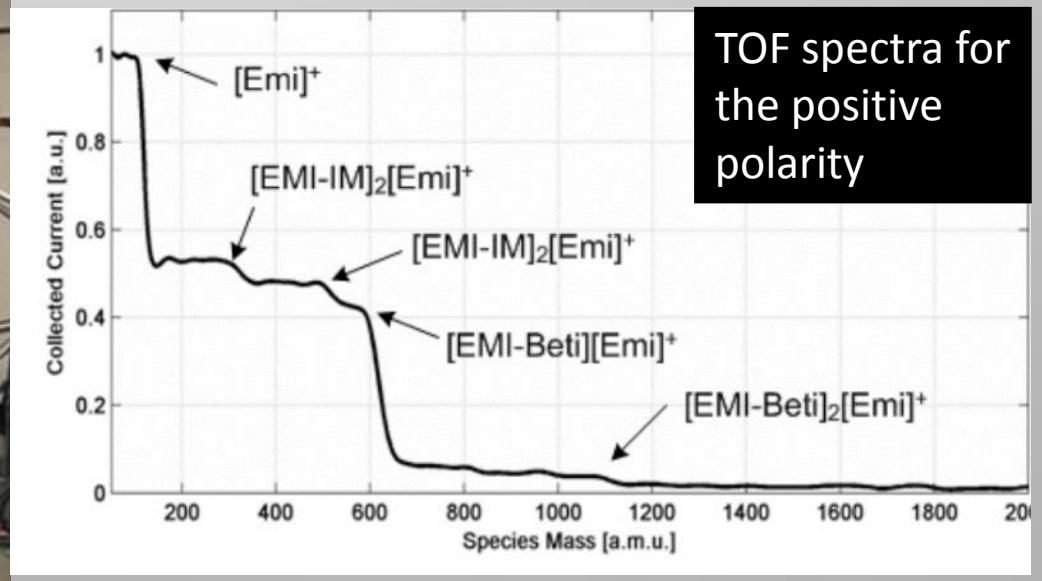

Same current on both polarities.

# Electronics

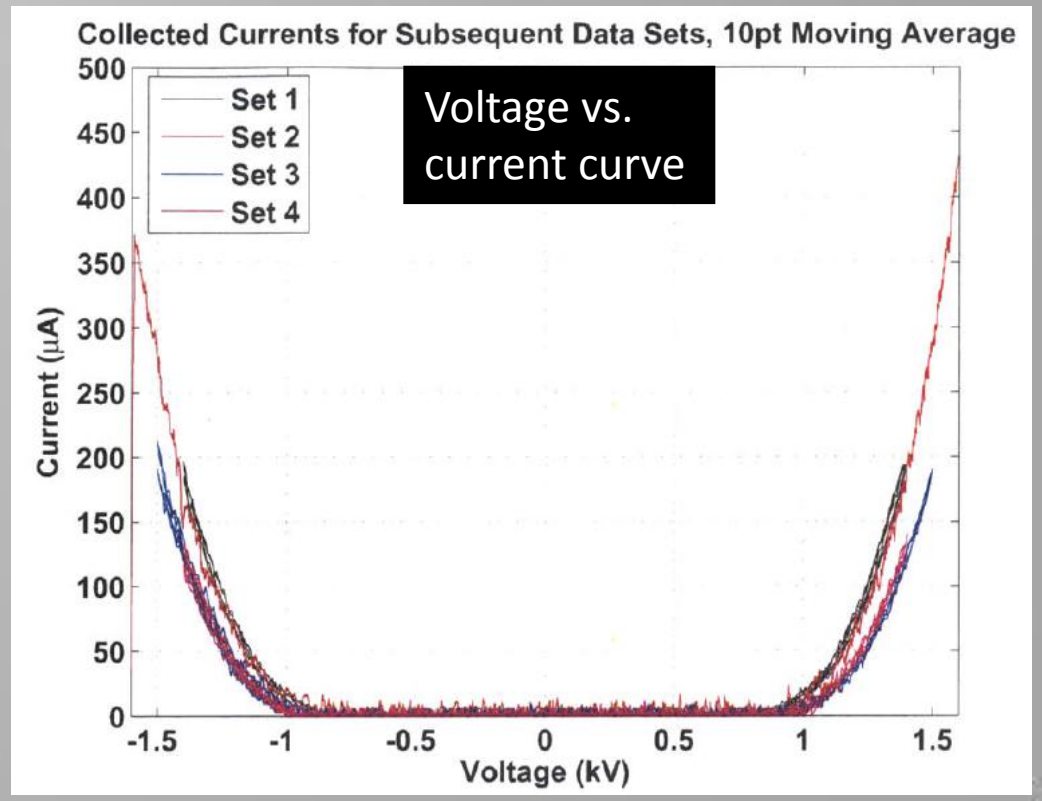
# Characterization of electrospray thrusters

- Voltage vs. current curves
- Time of flight (TOF) spectroscopy to characterize different ion species
- TRL level of 4-5 moving to 6 (Tsat)

Vacuum chamber at SPL

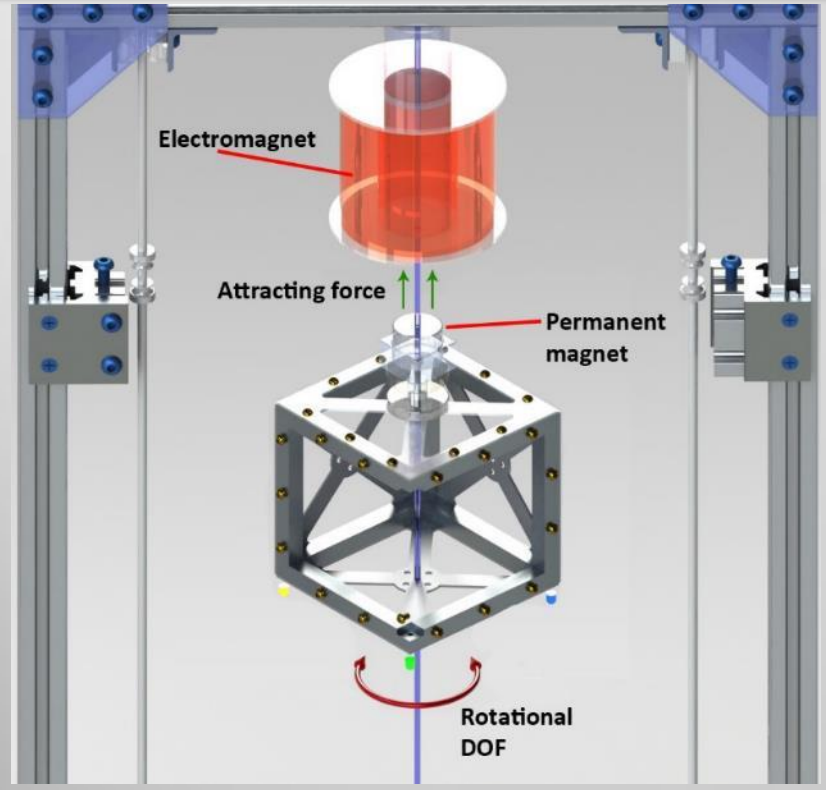


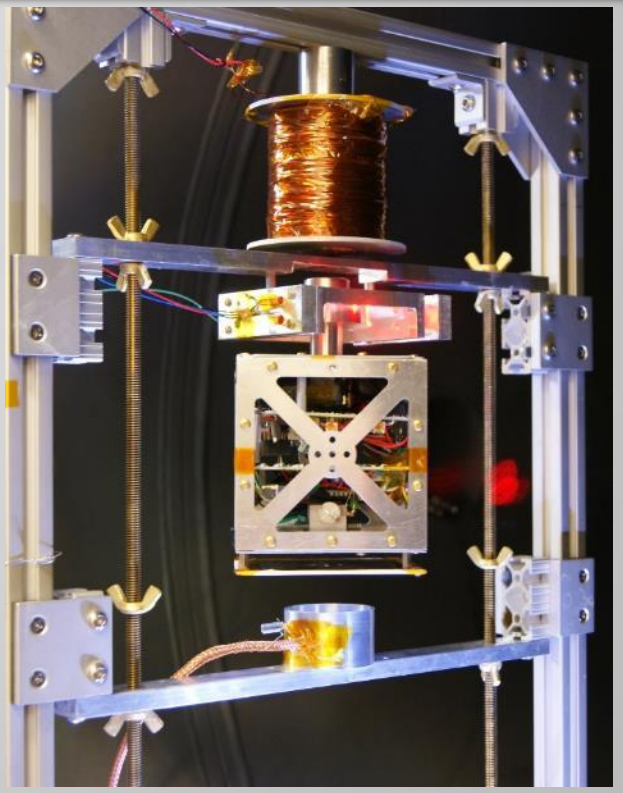

TOF spectra for the positive polarity

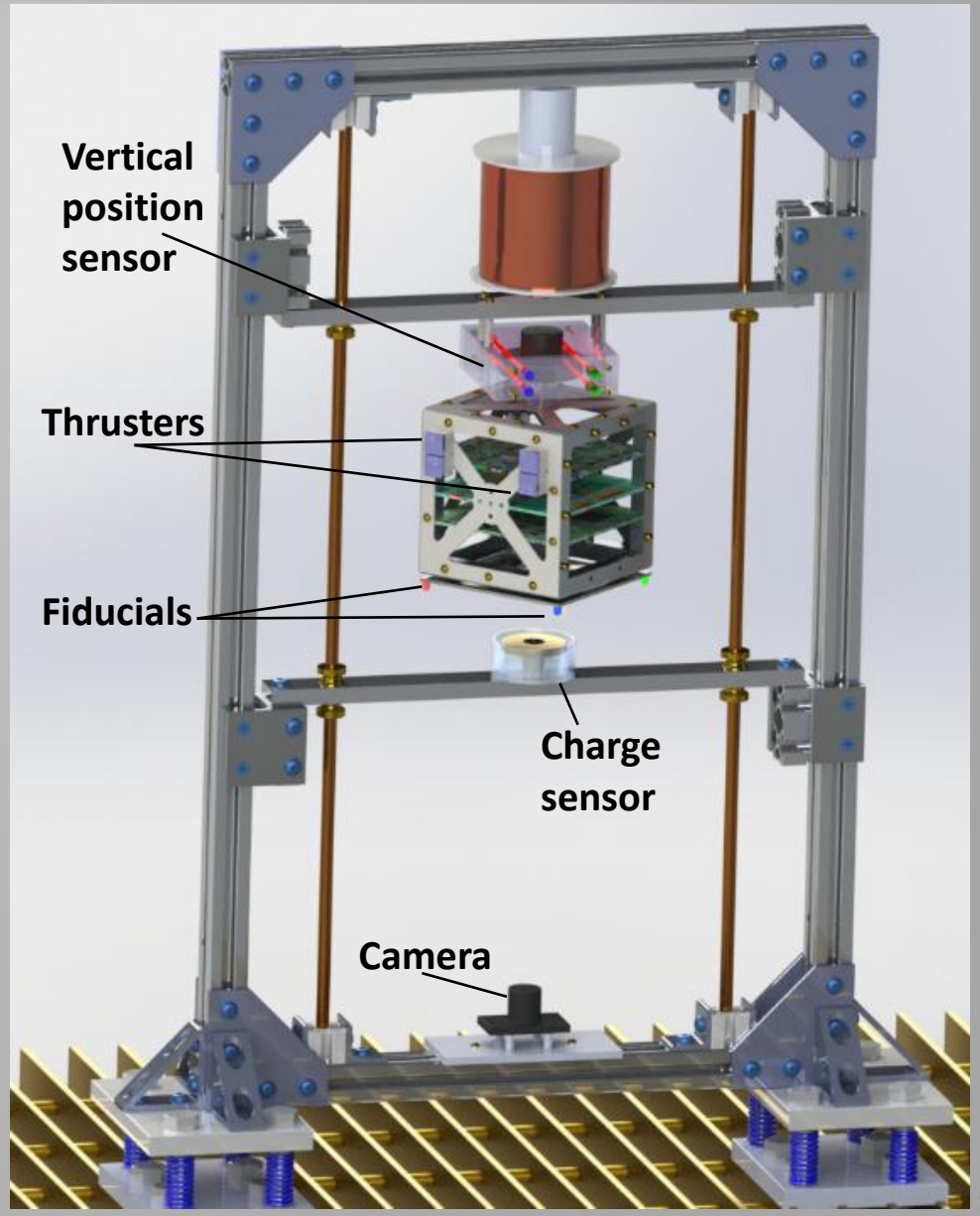



Voltage vs. current curve

# Testing, magnetically levitated thrust balance






- Angular position sensor (±1 degree)
- Vertical position range (±10 microns)
- Electrometer charge sensor
- Thrusters current and voltage levels
- Wireless link to PC

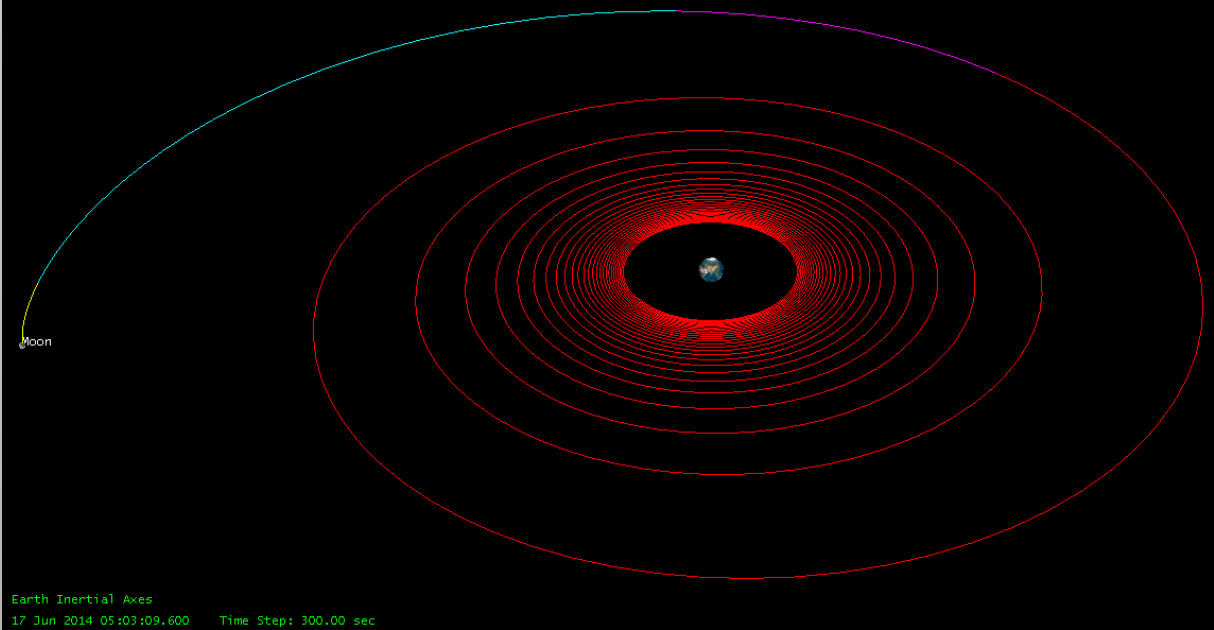
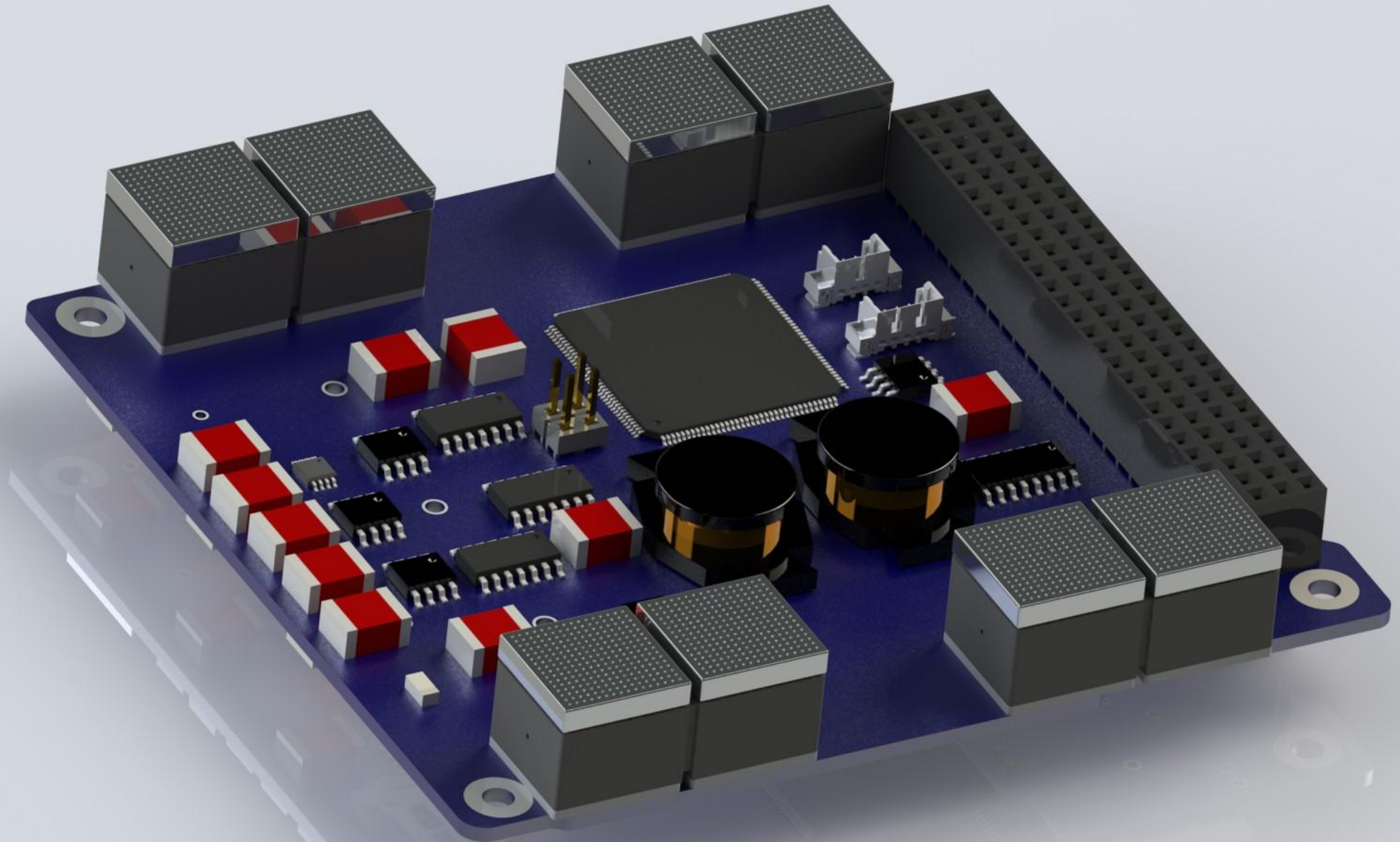
# Conclusions

- Attitude control and station keeping
- Orbit transfer, interplanetary transfers
- Formation flight
- Drag cancellation or active deorbit
- Precision pointing

| Mission examples assuming **1kg 1U** CubeSat with **1.5 Watts** available thrust power | | |
|---|---|---|
| **Maneuver** | **Time** | **Propellant** |
| Raise, lower orbit altitude by 400 km | 26 days | 6.4 grams |
| Spin spacecraft to 1 rev/sec | 1 hour | 10 milligrams |
| Walk in orbit 10 km | 5 hour | 51 milligrams |

3U CubeSat with 30 watts available thrust power going from GEO to moon.

- 100cm$^2$ thruster area, 16 µN/cm$^2$, 1.6 mN total thrust
- 3U CubeSat, 5kg
- ISP 2400
- 400 grams propellant
- 72 days constant burn



# Questions